

These additional input/output lines are used for tasks such as timing and coordinating between the data sender and the data recipient, so that the data does not get lost or abstruse while being transferred.

For robotics application, the data transfer rate need not be too high, and on a microcontroller, there is always scarcity of input/output ports. So, serial transmission is generally preferred in basic robotics applications. Although implementation of serial interface may seem difficult, it's not so if you use the proper software and hardware. The software works with the incoming serial data and the whole task of recombining the bit wise information into a byte, is performed by the processor. 